

ONE SITE, ONE HEAD: FOUR OPERATIONS

1. RETAIN
Old $S \rightarrow$ row-wise decay
 $\bar{S} = D(\alpha) S$

2. PREDICT
Read with current key
prediction = $\bar{S}^T k$

3. CORRECT
error = $v - \text{prediction}$
 $S = \bar{S} + \beta k \text{ error}^T$

4. READ
Read updated memory
output = $S^T q$

Key asks: what do I already predict for this association?
Query asks: what should downstream processing receive now?

The current write occurs before the current read. Keys and queries may differ.